

Problem 6 – Loss of prestress

The cross-section of a post-tensioned concrete beam spanning 16 m, simply supported, is shown in Figure 6.1. The beam is post-tensioned with a tendon, which has a parabolic profile, with end eccentricity zero and maximum eccentricity at midspan of 350 mm. The tendon comprises of 10 number 12.7 mm super grade low relaxation strands. Tendon is stressed to 70% of the breaking load. Material properties are:

Concrete:

$$f'_c = 32 \text{ MPa}$$

$$f'_{cp} = 25 \text{ MPa (at 14 days when the tendon is stressed)}$$

$$\text{specific weight of concrete} = 25 \text{ kN} / \text{m}^3$$

Prestressing Strands:

$$\text{Cross-sectional area of a single strand} = 100 \text{ mm}^2$$

$$\text{Yield strength of tendons, } f_{py} = 1640 \text{ MPa}$$

$$\text{Minimum breaking load of strands} = 184 \text{ kN}$$

$$E_p = 190,000 \text{ MPa}$$

Ordinary Reinforcement:

$$A_{st} = 8N24 = 3600 \text{ mm}^2, \text{ near coastal area}$$

Calculate loss of prestress.

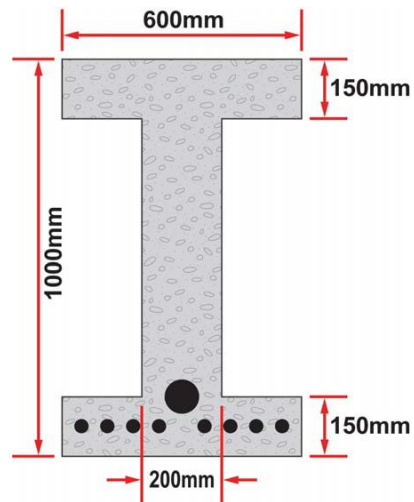


Figure 6.1 – Post-tensioned beam section